

[SCHOOL NAME — PLACEHOLDER LETTERHEAD]

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Class 4 Maths — CA-1 (Medium)

Class : 4

Subject : Maths (Math-Mela)

Max Marks : 70

Duration : 2 Hr 30 Min

Syllabus: Chapters 1–3 — Shapes Around Us, Hide and Seek, Pattern Around Us

General Instructions: All questions are compulsory. For Section A, write the correct option letter.

SECTION A — Multiple Choice Questions (1 Mark each)

1. A cuboid has 6 faces, 12 edges, and how many corners, following Faces + Corners - Edges = 2? (1 Mark)

A. 6

B. 8

C. 10

D. 12

2. Which of these shapes has NO straight sides or angles at all? (1 Mark)

A. Triangle

B. Square

C. Circle

D. Rectangle

3. If a shape has 6 sides, how many angles does it have? (1 Mark)

A. 4

B. 5

C. 6

D. 8

4. If Mini is looking at a building from a helicopter directly above it, which view is she seeing? (1 Mark)

A. Front view

B. Side view

C. Top view

D. None of these

5. On a grid, an object at row 3, column 1 is located _____. (1 Mark)

A. in the top-right corner

B. in the third row, first column

C. outside the grid

D. in the first row, third column

6. Which statement about odd and even numbers is TRUE? (1 Mark)

A. An odd number plus an odd number is always odd

B. An even number plus an even number is always even

C. All numbers are even

D. Odd numbers cannot be added

7. To pay exactly ₹75 using the fewest notes/coins, which combination works best (using 50, 20, 5)? (1 Mark)

A. 50 + 20 + 5

B. 20 + 20 + 20 + 5 + 5 + 5

C. 5 × 15

D. 50 + 5 × 5

8. A prism and a pyramid both have flat faces, but which of these is a key difference between them? (1 Mark)

A. A prism has two identical matching bases; a pyramid has only one base

B. A pyramid has more faces than any prism

C. Prisms cannot have rectangular faces

D. There is no difference between them

9. If a coin collection has 24 coins, can they always be grouped into complete pairs? (1 Mark)

A. Yes, because 24 is even

B. No, because 24 is odd

C. It depends on the coin size

D. Coins cannot be paired

SECTION B — Short Answer Questions (2 Marks each)

1. Explain how you would identify whether a 2D shape is a pentagon or a hexagon. (0 Marks)

2. A cube and a cuboid both have 6 faces, 12 edges, and 8 corners. Explain how you would tell them apart just by looking. (0 Marks)

3. Explain why the top view of a round object like a ball looks different from the top view of a cube. (0 Marks)

4. If you are standing at the school gate and a friend is on the rooftop looking down, explain why you would see two different views of the same building. (0 Marks)

5. Explain how a grid can help someone find a hidden object in a "treasure hunt" game. (0 Marks)

6. A group of 33 students wants to form pairs for a game. Explain what will happen and why. (0 Marks)

7. Explain two different ways to make a total of ₹60 using notes/coins of value 50, 20, 10, and 5. (0 Marks)

8. Why might builders use knowledge of prisms and pyramids when designing real buildings or monuments? (0 Marks)

9. Explain the pattern you would notice if you listed all odd numbers from 1 to 15. (0 Marks)

10. If a 2D shape has exactly 4 right angles and all four sides equal, what shape is it, and how do you know? (0 Marks)
11. Explain why a map of a classroom (top view) would not show the height of the furniture. (0 Marks)
12. A vendor sells items and always gives change using the fewest coins/notes possible. Explain why this is a practical strategy. (0 Marks)
13. Explain the difference between an edge and a face using a real object like a matchbox. (0 Marks)
14. Why might understanding grid positions be useful when playing a strategy game like chess or Battleship? (0 Marks)

SECTION C — Short Answer Questions (Explain) (3 Marks each)

1. Explain in detail how you would determine the number of faces, edges, and corners of a triangular prism, and describe what makes it different from a triangular pyramid. (0 Marks)
2. Describe a real-world scenario where understanding top, front, and side views would be genuinely useful, and explain why. (0 Marks)
3. Explain how patterns in even and odd numbers can help predict outcomes, using the example of pairing students for a group activity. (0 Marks)
4. A shopkeeper needs to give ₹235 in change using the fewest possible notes and coins (available: ₹200, ₹100, ₹50, ₹20, ₹10, ₹5). Explain the best combination and your reasoning. (0 Marks)
5. Explain how the relationship between the number of sides and angles in 2D shapes can help you identify an unknown shape quickly. (0 Marks)
6. Discuss why both 2D shape knowledge (Shapes Around Us) and spatial view knowledge (Hide and Seek) are useful together when reading a building blueprint. (0 Marks)

SECTION D — Long Answer Questions (5 Marks each)

1. Design a simple 3D model (like a small house or toy) using at least three different 3D shapes learnt in "Shapes Around Us." Describe each shape used, its properties (faces, edges, corners), and why you chose it for that part of your model. (0 Marks)
2. Explain in detail how a drone flying over your school, combined with a grid map of the classroom, could be used together to create a complete "guide" to your school for a new student. Describe what information each part (drone view and grid map) would provide. (0 Marks)
3. Explain, using detailed examples, how patterns (even/odd numbers, money combinations, and repeating sequences) appear in different parts of daily life, and discuss why recognising patterns is an important mathematical skill. (0 Marks)